

When Does Conformalization Help? A Cross-Paradigm Analysis of Probabilistic Electricity Price Forecasting

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Abstract

Probabilistic electricity price forecasting (EPF) relies on several uncertainty-quantification paradigms whose behavior under regime shift is not well understood. Conformalized Quantile Regression (CQR) guarantees coverage only when calibration and test data are exchangeable, which regime shift violates. We ask whether the base-model paradigm affects CQR's cost in the Continuous Ranked Probability Score (CRPS). On the Dutch EPEX day-ahead market across six origins spanning 2018–2024, we evaluate Quantile Regression (QR), LASSO-regularized QR (LQR-BIC), Quantile Regression Forests (QRF), quantile Light Gradient Boosting Machine (LGBM-Q), and Natural Gradient Boosting (NGBoost), each with and without CQR. Three findings emerge. The paradigms agree in calm regimes but linear methods lead in crisis. CQR is sharply paradigm-dependent: at the September 2021 regime shift, tree models incur 40–68% higher CRPS while linear models are unaffected or improve. And LQR-BIC+CQR achieves 13% lower CRPS than QR+CQR. We trace the mechanism to the differing residual-scale behavior of tree versus linear base models under regime shift and discuss its implications for regime-aware probabilistic forecasting practice.

CCS Concepts

• **Computing methodologies** → **Supervised learning by regression**; • **Mathematics of computing** → *Time series analysis*; *Probability and statistics*; • **Applied computing** → *Forecasting*.

Keywords

probabilistic forecasting, conformal prediction, quantile regression, electricity price forecasting, regime shift, cross-paradigm comparison

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1 Introduction

The European wholesale electricity market has undergone structural change since 2021 [8]. The 2021 gas-supply tightening and 2022 European energy crisis moved Dutch EPEX day-ahead prices from a pre-crisis median near 40 EUR/MWh, with hours rarely above 100, to sustained July–August 2022 stretches where more than 40% of hours exceeded 400 EUR/MWh, with the August peak above 800. Over the same period, negative prices during renewable oversupply grew from under 1% of hours before 2021 to more than 6% in 2024, extending the lower tail. The statistics here use the ENTSO-E data of Section 4.

For forecasters this poses a boundary problem periodic retraining does not fully solve: whenever a training window ends at or near a regime transition, its most recent slice, which serves as the conformal calibration set (Section 3.3), and the subsequent test window can straddle a structural break despite a fresh refit. The stakes are operational, since day-ahead prices drive battery dispatch, demand-response triggers, and hedging decisions. What matters is *calibrated* uncertainty, not merely point accuracy [7, 20], which concerns only the forecast's location and is silent about coverage.

Several probabilistic forecasting paradigms have become standard in EPF, and they differ along two axes that do not coincide. The first is the estimation target. Direct *quantile regression* (QR) [12] and the Light Gradient Boosting Machine (LightGBM) [11] with a pinball objective (LGBM-Q) target conditional quantiles directly, as does the ℓ_1 -regularized QR variant with the penalty selected by the quantile Bayesian Information Criterion (BIC), introduced to EPF for forecast pooling in [28] and applied here to raw covariates (LQR-BIC). Quantile Regression Forests (QRF) [19] estimates the conditional distribution nonparametrically from leaf-level empirical distributions, though the trees themselves are grown under a squared-error splitting rule. Natural Gradient Boosting (NGBoost) [6] instead fits the parameters of an assumed distribution under a proper scoring rule. The second axis is functional form, which sets the linear QR and LQR-BIC apart from QRF, LGBM-Q, and NGBoost, the last configured here with regression-tree base learners (Section 3.2). A recent review places all five within a broader uncertainty-quantification (UQ) taxonomy [25]. The two axes imply different extrapolation behavior beyond the training response support, as occurs under a regime transition, yet a cross-paradigm comparison on a single market with coverage tests under regime shift is, to our knowledge, absent.

The closest prior study [16] benchmarks distributional deep neural networks, based on ensembles, Monte Carlo dropout, and deep evidential regression, against XGBoost and LASSO-estimated autoregressive (LEAR) models combined with QRA and GARCH. On

the German market, over a single contiguous test year from December 2023 to November 2024, it applies split conformal prediction to the base models' point forecasts using absolute-residual nonconformity scores. Our study differs in three ways. The market is Dutch EPEX. The evaluation is structured as six anchor-based training–test episodes purposively spanning calm and crisis regimes between 2018 and 2024. The conformal method is CQR, whose nonconformity score is defined relative to the base model's predicted quantile boundaries rather than to point-forecast residuals (Section 3.3). CQR [24] is a distribution-free post-hoc wrapper for any quantile-producing base model, guaranteeing marginal coverage at least $1-\alpha$ provided calibration and test points are exchangeable. Exchangeability is weaker than independent and identically distributed (i.i.d.) sampling but fails outright when calibration and test sets straddle a regime boundary, and a recent line of non-exchangeable conformal prediction work has begun to characterize how distribution shift affects the coverage guarantee itself [2, 32, 33]. A separate question remains largely unexamined: even at a fixed level of exchangeability violation, does the *paradigm* of the base model, meaning its estimation target and functional form, systematically affect the downstream effect of CQR wrapping, as measured by proper scoring rules such as the Continuous Ranked Probability Score (CRPS)? This paper presents empirical evidence that it does. The contributions of this paper are twofold:

- *Empirical.* We evaluate five probabilistic EPF methods (QR, LQR-BIC, QRF, LGBM-Q, NGBoost), each with and without CQR, against a naive benchmark on the Dutch EPEX day-ahead market, across six anchor dates spanning 2018–2024. The anchors span four price regimes, from stable pre-crisis conditions through the 2021–2022 crisis to post-crisis stabilization (Section 4). We report point and probabilistic accuracy, coverage calibration, and formal coverage tests (Section 4).
- *Mechanistic.* We document a paradigm-conditional regime-shift interaction: at the canonical regime-shift anchor (September 2021), tree-based base models incur 40–68% higher CRPS under CQR while linear ones improve by 3.5–5%. We trace this asymmetry to the differing extrapolation of tree and linear quantile estimators and to the calibration set's location relative to the test-window regime. The observation motivates regime-stratified conformalization schemes whose theoretical analysis and empirical validation are deferred to future work.

2 Related Work

Probabilistic EPF. Review-level treatments include [15, 20, 30], with [20] focused on probabilistic forecasting and [15] curating the open-access benchmark now standard for the field. Probabilistic EPF has developed along three lines. The first, Quantile Regression Averaging [36], builds prediction intervals by fitting quantile regression to a pool of point forecasts as regressors, so the target's conditional quantiles are the output. Later work added automated variable selection and shrinkage for EPF regression [26], and, in [28], a LASSO-regularized QRA whose penalty is selected by a quantile BIC [17], a cheaper alternative to cross-validation. The second line treats forecasting as distributional estimation, in-

cluding EPF-specific distributional neural networks [18] and the more general Natural Gradient Boosting [6], which fits boosted base learners (in practice, decision trees) to the parameters of a conditional distribution. The third adapts tree-based methods directly to probabilistic output: Quantile Regression Forests [19], with recent production-ready implementations [9], and gradient-boosted trees [11] trained with the pinball (quantile) loss. These are widely used but have received limited formal analysis in EPF. A parallel line combines probabilistic forecasts by averaging distributional forecasts across training windows of different lengths [37].¹ Our evaluation places these on a common footing, cross-classified by the two axes of Section 1: linear (QR, LQR-BIC) against tree-based (QRF, LGBM-Q, NGBoost) in functional form, and direct-quantile (QR, LQR-BIC, LGBM-Q) against distributional (QRF nonparametric, NGBoost parametric) in estimation target.

Conformal prediction. Foundational treatments appear in [29], with a recent tutorial in [1]. Split conformal prediction with absolute-residual scores gives distribution-free finite-sample coverage but constant-width intervals, a limitation addressed by locally adaptive scores. CQR [24] combines quantile regression with split conformal calibration to yield input-dependent widths while retaining marginal coverage under exchangeability. Conditional coverage, by contrast, is provably impossible in finite samples without distributional assumptions [34], motivating weaker targets such as group-conditional coverage. The standard construction for it is Mondrian conformal prediction [29], which partitions the example space by a taxonomy and calibrates within each category, so the guarantee holds per category rather than only on average. Conformal prediction has seen growing adoption in EPF, beginning with [10]. The closest recent work [21] benchmarks time-series-adapted conformal schemes, including Ensemble Batch Prediction Intervals [38], against quantile regression on the Irish day-ahead and balancing markets, reporting that conformal interval width depends on the underlying regressor's accuracy. It compares several conformal schemes, each wrapped around multiple point-forecast regressors, over a single contiguous period. We take the complementary approach: fix the conformal method at CQR, vary the probabilistic base-model paradigm, and stress-test across structural breaks.

Non-exchangeable conformal prediction and distribution shift. CQR's marginal coverage guarantee assumes exchangeability, which fails when calibration and test samples come from different regimes. Recent theory characterizes the consequences: finite-sample bounds for non-exchangeable data are given in [2], with the coverage gap controlled by a weighted sum of total-variation distances measuring deviation from exchangeability. Adaptive conformal inference [32, 33] updates the effective calibration level online from realized miscoverage to hold the target coverage rate under arbitrary drift. This online approach has been applied to French spot prices [39], where offline split conformal loses coverage under temporal dependence and online recalibration restores it. For sequential data, ensemble-based CP [38] uses leave-one-out residuals from a bootstrap ensemble to avoid data splitting, attaining approximate finite-sample marginal coverage under strongly-mixing

¹What the EPF literature terms a *calibration window* [35, 37] is what we call a *training window*. See the terminology note at the top of Section 3.

errors, weaker than standard CP's exchangeability requirement. These methods give coverage guarantees of various strengths under drift, but all concern the conformal correction itself rather than the base model it wraps. Recent conformal-EPF work likewise either compares conformal methods on a single market [21], adapts calibration online under drift [39], or applies split conformal to the point forecasts of its base models [16]. To our knowledge, none isolates the paradigm-conditional cost of a fixed CQR correction across base-model families under a multi-anchor regime-shift stress test.

3 Methodology

Terminology. *Calibration* carries three meanings in the literatures we draw on, so we fix usage here. In EPF the *calibration window* is the historical training period used to fit a model [35, 37]. In split conformal prediction the *calibration set* $\mathcal{T}_{\text{cal}}$ is the held-out training portion on which conformity scores are computed [24]. In forecast evaluation, *coverage calibration* denotes the match between nominal and empirical interval coverage [7]. Throughout, *training window* ($\mathcal{T}_{\text{train}}$, 365 days) and *test window* ($\mathcal{T}_{\text{test}}$, 30 days) denote the chronological EPF split. The *calibration set* ($\mathcal{T}_{\text{cal}}$) refers only to the conformal held-out subset of $\mathcal{T}_{\text{train}}$. We reserve *coverage* for the property assessed by PICP and the Kupiec/Christoffersen tests.

3.1 Notation and Problem Setup

Let $Y_t \in \mathbb{R}$ denote the Dutch EPEX day-ahead clearing price at hour t . The feature vector $\mathbf{X}_t \in \mathbb{R}^d$ uses only data observable strictly before hour t , so no future information leaks into any prediction (per-model feature pools are given in Section 4). It contains (i) deterministic calendar features (including hour, day-of-week, day-of-year, month, weekend and peak-hour indicators) with their cyclical sine/cosine encodings; (ii) day-ahead forecasts for hour t (load, solar, onshore-wind, offshore-wind generation), issued by the system operator before delivery and therefore known before t ; (iii) autoregressive price lags Y_{t-k} for $k \in \{24, 48, 72, 168\}$; (iv) rolling statistics on strictly lagged price history, a previous-day block (min, max, mean, standard deviation over the trailing 24 hours, numerically coinciding with the 24-hour rolling statistics and retained from the baseline feature set), and the previous-day close, the price in the final hour of day $d-1$; (v) realized-volatility and vol-of-vol measures with squared and absolute returns, all on strictly lagged price history; (vi) commodity prices (TTF gas, EUA emissions, coal, oil), each shifted by 24 hours so that the value at hour t on day d is day $d-1$'s close; and (vii) interactions pairing renewable forecasts with calendar indicators (e.g., solar $\times$ afternoon, wind $\times$ night) and cyclical hour encodings, capturing duck-curve effects.

Given training data $\{(\mathbf{X}_t, Y_t)\}_{t \in \mathcal{T}_{\text{train}}}$, the goal is to predict, for each test hour $t \in \mathcal{T}_{\text{test}}$, the conditional quantiles of the price at the nine levels $\mathcal{A} = \{0.01, 0.05, 0.10, 0.25, 0.50, 0.75, 0.90, 0.95, 0.99\}$.² For a level $\alpha \in \mathcal{A}$, the estimation target is the true conditional α -quantile $Q_\alpha(\mathbf{x}) := \inf\{y \in \mathbb{R} : \Pr(Y \leq y \mid \mathbf{X} = \mathbf{x}) \geq \alpha\}$, the level the price falls below with probability α given the features. Each base model returns an estimator $\hat{Q}_\alpha : \mathbb{R}^d \rightarrow \mathbb{R}$ of Q_α , reached in three ways. QR, LQR-BIC, and LGBM-Q minimize the pinball

loss $\rho_\alpha(u) = u(\alpha - \mathbf{1}_{u < 0})$, averaged over the training data at the residual $u = Y_t - \hat{Q}_\alpha(\mathbf{X}_t)$, where $\mathbf{1}_{u < 0}$ indicates the event $u < 0$. It weights under-prediction by α and over-prediction by $1 - \alpha$, and its population minimizer is exactly Q_α [7, 12]. NGBoost fits conditional distribution parameters by a proper scoring rule and reads quantiles off it. QRF grows trees with the standard regression splitting criterion and estimates quantiles from the conditional empirical distribution induced by the forest weights [19]. Section 3.2 gives details.

At hour t the nine estimates $\{\hat{Q}_\alpha(\mathbf{X}_t)\}_{\alpha \in \mathcal{A}}$ form the predictive quantile function CQR later modifies. Reported CRPS values use trapezoidal integration of the pinball-CRPS identity over this grid. We follow a fit-once-per-anchor protocol: parameters are estimated once on $\mathcal{T}_{\text{train}}$ and frozen throughout $\mathcal{T}_{\text{test}}$, while $\mathbf{X}_t$ uses only information available before delivery day d , enforced by the lag structure above. Section 4 gives the full protocol. When CQR is applied, $\mathcal{T}_{\text{train}}$ is further partitioned into $\mathcal{T}_{\text{proper}}$ and $\mathcal{T}_{\text{cal}}$, defined in Section 3.3. Hyperparameter tuning (Section 4) uses all of $\mathcal{T}_{\text{train}}$ and is shared between each base model and its CQR-wrapped counterpart, so any CQR-induced change is attributable to the conformalization step rather than retuning.

3.2 Five Probabilistic Methods and a Naive Benchmark

Except for the naive benchmark, QR, and LQR-BIC, hyperparameters are tuned via 5-fold purged time-series cross-validation [22] with CRPS as the objective; the searched grids are given in Section 4.

Naive benchmark. We adapt the day-type lag rule of [20]: the forecast at hour h of day d is $Y_{h,d-7}$ for Mondays, Saturdays, and Sundays, and $Y_{h,d-1}$ for Tuesdays–Fridays. In the original formulation the lagged price is observed. We instead apply it recursively across the 30-day test window without ground-truth feedback, initialized from the trailing 14 days of training prices, so prior naive forecasts substitute for unobserved actuals once the horizon exceeds the active lag. This is a stricter benchmark than the observed-lag formulation, and the rMAE values in Section 5 should be read accordingly. Intervals come from the empirical distribution of in-sample residuals $r_t = Y_t - Y_{t-\text{lag}(t)}$ on actual training prices, where $\text{lag}(t) \in \{24, 168\}$ depends on the weekday of t .

Quantile Regression (QR). For each $\alpha \in \mathcal{A}$ we solve

$$\hat{\beta}^\alpha = \arg \min_{\beta} \sum_{t \in \mathcal{T}_{\text{train}}} \rho_\alpha(Y_t - \mathbf{X}_t^\top \beta), \quad (1)$$

where $\beta \in \mathbb{R}^d$ is the coefficient vector of the linear quantile function, fitted separately at each α to give $\hat{\beta}^\alpha$, and an intercept is absorbed into $\mathbf{X}_t$. Under mild regularity and a correctly specified linear conditional quantile function, $\hat{\beta}^\alpha$ is $\sqrt{n}$ -consistent and asymptotically normal [12, 13]. The inductive bias is linearity, so the fitted function is restricted to hyperplanes in $\mathbf{X}_t$ and extrapolation beyond the training support follows the fitted coefficient structure, a property that matters under regime shift.

LASSO-regularized QR with BIC (LQR-BIC). Adapting the BIC-tuned LASSO quantile-regression framework of [28], whose regressors are a pool of point forecasts, to our raw-covariate setting, each quantile estimator adds an ℓ_1 penalty

²Throughout this section α denotes a quantile level. In the CQR construction of Section 3.3 it denotes a miscoverage rate.

$$\hat{\beta}_\lambda^\alpha = \arg \min_{\beta} \frac{1}{n} \sum_{t \in \mathcal{T}_{\text{train}}} \rho_\alpha(Y_t - \mathbf{X}_t^\top \beta) + \lambda \|\beta\|_1, \quad (2)$$

where $\lambda \geq 0$ is the regularization parameter and $n = |\mathcal{T}_{\text{train}}|$ the training sample size. It is selected, separately at each α , by minimizing a quantile-analogue BIC over a finite grid,

$$\text{BIC}(\lambda) = \log \left(\sum_{t \in \mathcal{T}_{\text{train}}} \rho_\alpha(Y_t - \mathbf{X}_t^\top \hat{\beta}_\lambda^\alpha) \right) + \frac{m_\lambda \log n \log d}{2n}, \quad (3)$$

where m_λ is the number of non-zero entries of $\hat{\beta}_\lambda^\alpha$ and d is the feature dimension. This is the quantile BIC of [17], adopted for EPF by [28]. LQR-BIC shares QR's linear inductive bias while adding sparsity.

Quantile Regression Forests (QRF). We use the implementation of [9], which retains leaf-level response distributions and extracts quantiles at prediction time [19]. The inductive bias is locally constant estimation: at a test point the conditional response distribution is a weighted empirical distribution over training responses, each weight being the average over all trees of its inverse leaf size, zero on trees where it does not share a leaf with the test point. The α -quantile prediction is the corresponding empirical quantile. Relevant here, QRF cannot extrapolate beyond the training response range, since its predictions are quantiles of a distribution placed on training responses.

Quantile-LightGBM (LGBM-Q). A LightGBM [11] estimator is fitted per target quantile with pinball loss, so nine independent boosting models produce our quantile grid. Because each α -model is fit independently, the predictions can violate $\hat{Q}_\alpha(\mathbf{x}) \leq \hat{Q}_{\alpha'}(\mathbf{x})$ for $\alpha < \alpha'$, known as quantile crossing, corrected as described in Section 3.3. Monotonicity restoration yields valid quantile vectors but cannot fix the underlying issue that separate fits do not share leaf structure.

NGBoost. NGBoost [6] treats the parameters of an assumed conditional distribution as gradient-boosting targets, with the natural gradient correcting for reparametrization effects. Our configuration uses decision-tree base learners, a Normal distribution, and the logarithmic score, giving a conditional $(\hat{\mu}_t, \hat{\sigma}_t)$ pair at each test point from which all quantiles follow analytically. The inductive bias combines tree-based estimation of conditional location and scale with a globally fixed parametric family.

3.3 Conformalized Quantile Regression (CQR)

Given a base model producing quantile estimates $\hat{Q}_\alpha$, *symmetric* CQR [24, Algorithm 1] proceeds as follows. Fix a miscoverage level $\alpha \in (0, 1)$ and set $\alpha_{\text{lo}} = \alpha/2$, $\alpha_{\text{hi}} = 1 - \alpha/2$. Partition the training set into $\mathcal{T}_{\text{proper}}$ and $\mathcal{T}_{\text{cal}}$, refit the base model on $\mathcal{T}_{\text{proper}}$ only, and compute on $\mathcal{T}_{\text{cal}}$ the *conformity scores*

$$S_t = \max \left(\hat{Q}_{\alpha_{\text{lo}}}(\mathbf{X}_t) - Y_t, Y_t - \hat{Q}_{\alpha_{\text{hi}}}(\mathbf{X}_t) \right), \quad t \in \mathcal{T}_{\text{cal}}. \quad (4)$$

Let $\hat{q}_{1-\alpha}$ denote the $\lceil (1-\alpha)(|\mathcal{T}_{\text{cal}}|+1) \rceil$ -th order statistic of $\{S_t\}_{t \in \mathcal{T}_{\text{cal}}}$, equivalently the empirical $\lceil (1-\alpha)(|\mathcal{T}_{\text{cal}}|+1) \rceil / |\mathcal{T}_{\text{cal}}|$ -quantile under the convention $\hat{F}^{-1}(p) = \inf\{x : \hat{F}(x) \geq p\}$, as required by [24, Theorem 1] for the coverage statement below.³ The lowercase distinguishes the scalar correction from the trained quantile function

³Our implementation uses `numpy.quantile` with linear interpolation, which returns a value no smaller than the order statistic, so the conformal correction is at least as large as the specification of [24, Theorem 1] and marginal coverage still holds.

$\hat{Q}_\alpha(\cdot)$. The conformalized interval at $\mathbf{x}$ is

$$\left[\hat{Q}_{\alpha_{\text{lo}}}(\mathbf{x}) - \hat{q}_{1-\alpha}, \hat{Q}_{\alpha_{\text{hi}}}(\mathbf{x}) + \hat{q}_{1-\alpha} \right]. \quad (5)$$

Under exchangeability of calibration and test pairs, this interval has coverage at least $1 - \alpha$ [24]. Two violations of that assumption operate at different orders here. Serial dependence in hourly prices voids strict exchangeability at every anchor, but under near-stationarity the calibration scores remain marginally representative of the test scores. The coverage gap is controlled by the deviation from exchangeability [2], and surfaces as temporal clustering of violations rather than marginal miscoverage (Section 5.1). Regime shift, by contrast, separates the calibration-score distribution from the test-score distribution itself. It is the first-order violation and the object of this paper, and one identified in [10] as breaking exchangeability without empirical testing. The effect on CRPS is non-trivial because Equation (5) can make intervals either wider or narrower than the base-model output. The asymmetric variant of [24, Theorem 2], which uses separate one-sided scores $E_t^{\text{lo}} = \hat{Q}_{\alpha_{\text{lo}}}(\mathbf{X}_t) - Y_t$ and $E_t^{\text{hi}} = Y_t - \hat{Q}_{\alpha_{\text{hi}}}(\mathbf{X}_t)$ with independent corrections per side, is left to future work.

We apply CQR to all five methods, taking $\mathcal{T}_{\text{cal}}$ as the chronologically last 15% of $\mathcal{T}_{\text{train}}$, which preserves temporal order and places the calibration set adjacent to the test window. This maximizes representativeness under near-stationary conditions, while exposing the correction to regime heterogeneity precisely when the end of training and the test window differ in regime (Section 5.2).

We instantiate CQR at two miscoverage levels, $\alpha \in \{0.5, 0.1\}$, calibrating the central pairs $(\hat{Q}_{0.25}, \hat{Q}_{0.75})$ and $(\hat{Q}_{0.05}, \hat{Q}_{0.95})$. The remaining levels, $\{0.01, 0.10, 0.50, 0.90, 0.99\}$, come from the refit base model uncorrected, so the reported CQR-wrapped CRPS aggregates four conformalized and five pass-through levels by trapezoidal integration⁴, consistent with conformal prediction practice in EPF where the correction is applied at the prediction-interval boundaries actually reported [10, 16].

4 Experimental Setup

4.1 Dataset

Hourly Dutch EPEX day-ahead data are drawn from the ENTSO-E Transparency Platform, spanning 2 January 2016 to 31 August 2024 (75,960 observations). The target is the day-ahead clearing price in EUR/MWh. The auction clears once per day at gate closure around midday on the day before delivery [15], at 12:00 CET in the Dutch market, but settles 24 hourly prices, one per hour of the delivery day, so the modeling target is hourly. The persistent intraday profile, with overnight troughs, a midday solar-driven dip, and an evening ramp, makes hour-of-day and its cyclical encoding essential covariates. Consistent with Section 3's requirement that $\mathbf{X}_t$ use only information observable strictly before hour t , the exogenous features are day-ahead load forecasts (not realized loads, observed

⁴Quantile levels fitted independently can yield a non-monotone sequence in α , and per-level CQR offsets can compound this. Because trapezoidal integration of pinball losses assumes a monotone quantile function [7], we restore monotonicity by row-wise sorting, the monotone rearrangement of [3], which does not increase the L_p distance to the true quantile function. It is applied to every model's predicted quantile sequence and again after the conformal offsets, and is a no-op for the monotone-by-construction methods.

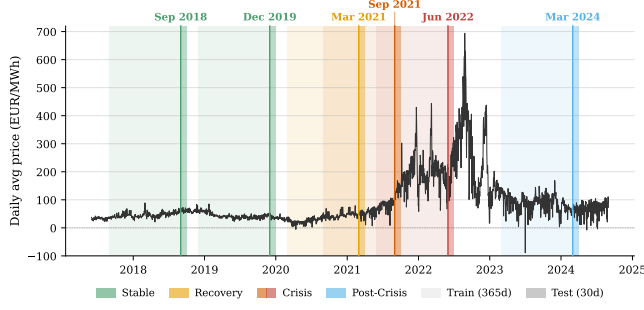


Figure 1: Daily average day-ahead price on the Dutch EPEX market with the six anchors. For each anchor the lighter band is its 365-day training window and the darker band its 30-day test window, with color encoding the regime of Table 1.

only at delivery and therefore unknown at gate closure), renewable-generation forecasts (solar, onshore wind, offshore wind), and commodity prices (TTF natural gas, API2 coal, EUA carbon, Brent oil). Commodity prices are daily series, forward-filled from the most recent close [18] to cover weekends and holidays. The value on delivery day d is the close of day $d-1$, which precedes delivery but follows the midday gate closure on $d-1$: this satisfies the measurability requirement above and is weaker than the two-day lag of [18]. Feature engineering produces 69 candidate predictors per anchor, comprising groups (i)–(vii) of Section 3 together with the renewable total, where rolling statistics use the 24h and 168h windows.

4.2 Multi-Anchor Evaluation Protocol

An *anchor* (or forecast origin) separates historical training data from an out-of-sample test period. The six were chosen purposively from the European energy event record and the realized price path, spanning calm, transitional, and crisis conditions. Two properties of an anchor matter, and they do not coincide.

The first is the price regime the test window falls in, which we label descriptively:

- **Stable:** 1 September 2018, 1 December 2019.
- **Recovery** (moderate upward drift after the 2020 demand collapse): 1 March 2021.
- **Crisis** (test window at sustained crisis price levels): 1 September 2021, the onset of the gas-supply tightening, and 1 June 2022, deep in the crisis period. We call these the *crisis anchors*.
- **Post-crisis** (downward normalization): 1 March 2024.

Figure 1 shows the daily average price path with all six anchors and their windows.

The second is how $\mathcal{T}_{\text{cal}}$ relates to $\mathcal{T}_{\text{test}}$, since the conformity scores of Section 3.3 are computed on the former and applied to the latter. Table 1 gives the median price in each window. Only September 2021 combines crisis prices with a large calibration-to-test shift: the median rises from 84.4 in calibration to 132.5 EUR/MWh in test, with 29.4% of test hours above the calibration maximum. At June 2022 the calibration set already sits at crisis levels (196 against 207), so calibration and test are matched despite the crisis. March 2024 differs again: its calibration-to-test ratio is unremarkable and matches the stable December 2019 anchor (0.93 in both), but the

Table 1: Anchor selection and regime classification. Median hourly price (EUR/MWh) in $\mathcal{T}_{\text{train}}$, the conformal calibration set $\mathcal{T}_{\text{cal}}$ (last 15% of training, Section 3.3), and $\mathcal{T}_{\text{test}}$. The ratios summarize the shift to the test window from the fitting sample and from the calibration sample. September 2021 is the only anchor combining crisis price levels with a large calibration-to-test shift.

Anchor	Regime	Median price			Ratio		% test > max $\mathcal{T}_{\text{cal}}$
		$\mathcal{T}_{\text{train}}$	$\mathcal{T}_{\text{cal}}$	$\mathcal{T}_{\text{test}}$	$\frac{\text{test}_{\text{train}}}{\text{test}_{\text{cal}}}$	$\frac{\text{test}_{\text{cal}}}{\text{test}_{\text{cal}}}$	
Sep 2018	Stable	44.5	55.1	62.3	1.40	1.13	1.8
Dec 2019	Stable	41.0	39.4	36.6	0.89	0.93	0.0
Mar 2021	Recovery	34.4	49.1	48.9	1.42	1.00	0.0
Sep 2021	Crisis	52.2	84.4	132.5	2.54	1.57	29.4
Jun 2022	Crisis	163.6	196.0	207.0	1.27	1.06	8.5
Mar 2024	Post-crisis	90.4	71.0	66.2	0.73	0.93	0.4

level declines steadily across the window, from a training median of 90.4 through 71 to 66.2, the largest downward move of the six. We call September 2021 and March 2024 the *regime-shift anchors*, the first an abrupt upward break, the second a sustained downward drift.

For each anchor, $\mathcal{T}_{\text{train}}$ is 365 days and $\mathcal{T}_{\text{test}}$ is 30 days. The 365-day length is a standard EPF choice whose sensitivity to alternative lengths (termed *calibration window lengths* in that literature; cf. Section 3) is studied systematically in [35]. Prices enter all models on the raw EUR/MWh scale, with no variance-stabilizing or other target transformation, keeping the tree-method extrapolation mechanism of Section 5.2 and the additive correction of Equation (5) interpretable in price units. This is also deliberate given the fit-once protocol: under a frozen per-anchor fit, a transform such as the scaled inverse-hyperbolic-sine [27] is estimated on the training regime and, at an upward regime boundary, maps linear extrapolation through its exponential inverse tail, so we defer such transforms to the rolling-retraining setting of future work. Missing values are median-imputed and features robust-standardized per anchor, in both cases using training-window statistics.

Forecast protocol. Each base model is fit once on $\mathcal{T}_{\text{train}}$ and each CQR variant once on $\mathcal{T}_{\text{proper}}$. The fitted model then produces all $30 \times 24 = 720$ hourly quantile forecasts over $\mathcal{T}_{\text{test}}$, with no parameter updates during the test window. The features $\mathbf{X}_t$ do vary across it, since each delivery day brings new lagged inputs, but obey a strict day-ahead constraint: every feature derived from the electricity price uses values settled at or before the auction preceding the delivery day, and no feature uses information from the delivery day itself, and the load and renewable inputs are day-ahead forecasts published before delivery. The protocol is therefore a once-daily block forecast of all 24 hours, not an hour-by-hour recursive forecast drawing on prices from within the day. This departs deliberately from the daily rolling re-estimation standard in operational EPF [15, 28, 35], for two reasons. First, freezing the fit isolates conformalization from retraining: a base model and its CQR wrapper share hyperparameters and feature pool and neither is updated within the test window, so any CRPS change between them is attributable to the conformalization step of Section 3.3, the proper-train/calibration split with the conformal correction, not to retuning. Second, and more central, the frozen calibration set is what exposes the exchangeability violation we study. Under daily

Table 2: Mean performance across six anchors. The best value in each column is shown in bold and the second-best is underlined. When two methods tie for the best value, both are shown in bold. Lower values indicate better performance for MAE, RMSE, CRPS, and Winkler-90. PICP-50 and PICP-90 should be close to 0.50 and 0.90, respectively. PINAW-90 is informative only jointly with PICP-90 and carries no emphasis.

Model	MAE	RMSE	CRPS	PICP-50	PICP-90	Winkler-90	PINAW-90
Naive	24.25	31.31	18.05	0.34	0.76	177.98	0.40
QR	<u>14.11</u>	19.21	<u>9.85</u>	0.45	0.86	82.17	0.26
LQR-BIC	13.53	18.60	9.68	0.41	<u>0.84</u>	<u>87.13</u>	0.24
QRF	15.71	21.83	11.38	0.50	0.86	114.44	0.28
LGBM-Q	15.24	21.02	11.96	0.33	0.69	138.37	0.19
NGBoost	15.81	21.84	11.64	0.44	0.78	119.30	0.22
QR+CQR	16.17	20.91	11.98	<u>0.49</u>	0.80	129.13	0.28
LQR-BIC+CQR	14.29	<u>19.08</u>	10.41	0.44	0.82	95.44	0.26
QRF+CQR	19.12	25.16	14.33	0.39	0.81	151.94	0.30
LGBM-Q+CQR	19.06	25.11	14.84	0.35	0.71	182.39	0.24
NGBoost+CQR	19.19	25.09	14.52	0.35	0.76	150.50	0.24

rolling it slides forward and re-enters the prevailing regime within days, continuously repairing the calibration-versus-test mismatch. Freezing holds it in the pre-shift regime across the whole test window, precisely the configuration in which the paradigm-conditional CQR effect of Section 5.2 appears. Freezing is thus not a convenience but the instrument that surfaces the effect. A rolling-retraining variant, under which we expect attenuation as the calibration set tracks the regime, is the key robustness question deferred to future work (Section 5.4).

4.3 Cross-Validation and Feature Selection

Hyperparameters are tuned with 5-fold rolling-origin time-series cross-validation, a 7-day purge between each training fold and its validation fold [22], and CRPS scoring. Feature handling is paradigm-specific. QR uses unregularized pinball-loss minimization, which is unstable at high feature-to-sample ratios [13], so we pre-select 30 features per anchor by Pearson correlation with the target. The other four absorb the full 69-feature pool: LQR-BIC selects via ℓ_1 shrinkage with λ tuned per quantile by Equation (3). QRF and LGBM-Q are invariant to collinearity by construction, while NGBoost absorbs the pool inside the CV folds and in the final fit, relying on the implicit regularization of gradient boosting. We apply no variance-inflation-factor (VIF) pre-filtering because the asymmetric handling above already addresses QR, the only paradigm vulnerable to multicollinearity. The searched grids are: QRF, $n_estimators \in \{500, 800, 1000, 1500\}$, $min_samples_leaf \in \{2, 5, 10\}$, $max_features \in \{sqrt, 0.5, 0.7\}$, and $max_depth \in \{None, 15, 25\}$; LGBM-Q, $n_estimators \in \{100, 200, 300\}$, $learning_rate \in \{0.03, 0.05\}$, $max_depth \in \{6, 8\}$, $num_leaves \in \{15, 31, 63\}$, $min_child_samples \in \{10, 20, 50\}$, $reg_alpha \in \{0, 0.1, 1.0\}$, and $reg_lambda \in \{0, 0.1, 1.0\}$; NGBoost, $n_estimators \in \{500, 800, 1200\}$, $learning_rate \in \{0.003, 0.005, 0.01\}$, $max_depth \in \{5, 6, 7\}$, and $minibatch_frac \in \{0.7, 0.9\}$, with a Normal predictive distribution. The LQR-BIC penalty grid is $\lambda \in \{0, 0.001, 0.01, 0.1, 0.5, 1, 2, 5, 10, 20, 50\}$.

4.4 Evaluation Metrics

For point accuracy we report MAE and RMSE, together with three relative summaries that ease cross-study comparison: relative MAE

Table 3: Relative point-accuracy summaries pooled over all six test windows (4,320 hourly forecasts). rMAE is the MAE relative to the naive benchmark, so values below 1 indicate an improvement over the benchmark. CVRMSE and R^2 are for cross-study comparison only, and are not comparable across regimes (see text).

Model	rMAE	CVRMSE (%)	R^2
Naive	1.0000	42.0971	0.7084
QR	<u>0.5818</u>	25.2083	0.8955
LQR-BIC	0.5580	24.2520	0.9032
QRF	0.6477	28.1442	0.8697
LGBM-Q	0.6284	27.2498	0.8778
NGBoost	0.6520	28.3260	0.8680
QR+CQR	0.6669	29.1273	0.8604
LQR-BIC+CQR	0.5890	<u>25.1816</u>	<u>0.8957</u>
QRF+CQR	0.7886	32.9564	0.8213
LGBM-Q+CQR	0.7858	33.3408	0.8171
NGBoost+CQR	0.7913	33.5443	0.8149

(rMAE), the coefficient of variation of RMSE (CVRMSE), and R^2 . Here rMAE divides the model MAE by the naive-benchmark MAE, and CVRMSE divides RMSE by the mean realized price. Probabilistic accuracy rests on the CRPS [7] and the Winkler score [31]. To assess coverage calibration we use the Prediction Interval Coverage Probability (PICP) at 50% and 90% alongside the Prediction Interval Normalized Average Width (PINAW), which normalizes the mean interval width by the realized test-window price range [23]. Three coverage tests follow: Kupiec unconditional coverage [14], Christoffersen independence [4], and Ljung–Box residual autocorrelation at lags 24 and 168, all at $\alpha = 0.05$. Pairwise accuracy differences use the two-sided Diebold–Mariano test [5] on per-hour CRPS loss differentials, with a Newey–West long-run variance, Bartlett kernel, and truncation lag $\lceil n^{1/3} \rceil = 9$ for $n = 720$ test hours.

5 Results and Discussion

5.1 Aggregate Cross-Paradigm Performance

Table 2 summarizes mean performance across the six anchors.⁵ Linear methods are the most accurate on average: LQR-BIC attains both the lowest MAE (13.53) and the lowest CRPS (9.68), ahead of QR (9.85). Tree-based paradigms cluster at CRPS 11.38–11.96 (NGBoost 11.64), an 18–24% gap consistent with the EPF benchmark literature [15, 18]. LGBM-Q has the poorest marginal 90% coverage (0.69), with intervals roughly a quarter narrower (PINAW-90 = 0.19) than QR (0.26), which explains the under-coverage rather than indicating superior sharpness.

On rMAE (Table 3), every probabilistic method beats the naive benchmark by a wide margin, from 0.56 (LQR-BIC) to 0.65 (NGBoost), and conformalization raises rMAE in every case, consistent with the CRPS results. CVRMSE and R^2 require care under regime shift, since both normalize by the price level or variance, which jump across regimes, so they understate crisis degradation. Within QR, CVRMSE is 22% in calm and 21% in crisis, and R^2 is higher in crisis (0.74) than calm (0.69) even though the absolute crisis RMSE is about three times larger. Pooling inflates R^2 further, to 0.90, because

⁵Code and data: <https://github.com/ctzag/setn2026-cqr-epf>.

Table 4: Fraction of 6 anchors passing the Kupiec unconditional-coverage test at $\alpha = 0.05$.

Model	Kupiec-50 pass	Kupiec-90 pass
Naive	0/6	0/6
QR	1/6	1/6
QR+CQR	2/6	2/6
LQR-BIC	3/6	1/6
LQR-BIC+CQR	1/6	1/6
QRF	2/6	3/6
QRF+CQR	1/6	3/6
LGBM-Q	0/6	0/6
LGBM-Q+CQR	1/6	0/6
NGBoost	2/6	2/6
NGBoost+CQR	1/6	1/6

the model also tracks the between-regime level shift. The regime comparison therefore rests on absolute CRPS (Section 5.2). Only rMAE is a relative summary stable across regimes, its benchmark being exposed to the same shift.

Aggregate coverage calibration is mixed. Table 4 gives Kupiec unconditional-coverage pass rates. The strongest are LQR-BIC (3/6 at the 50% level) and QRF (3/6 at 90%). CQR-wrapped variants reach at most 3/6 (QRF+CQR at 90%), confirming that conformalization does not uniformly improve coverage when calibration and test windows differ in regime. Figure 2 plots PICP-90 against the nominal 0.90 target: trajectories vary substantially with anchor date, the two crisis anchors (Sep 2021, Jun 2022, shaded) producing the largest excursions below nominal in both panels. The Christoffersen independence and Ljung–Box residual-autocorrelation tests are rejected for every (model, anchor) pair,⁶ so interval violations cluster in time for every paradigm.

A complementary diagnostic is feature-selection consistency for QR, the only method using correlation-based pre-selection. The features selected at all six anchors are exactly those EPF domain knowledge predicts: price lags at 24, 48, 72, and 168 hours, the trailing 24-hour minimum, maximum, and mean (each through both duplicate encodings, cf. Section 4), the 168-hour rolling mean, the previous-day close, and the TTF gas price. That the same predictors survive substantial regime variation is reassuring: the inductive bias is selecting real market structure, not anchor-specific artifacts.

5.2 Regime Effects and Paradigm-Conditional CQR Degradation

Averaging across anchors hides a strong regime dependence. Table 5 splits mean CRPS by crisis (September 2021, June 2022) against non-crisis regimes. Away from crisis the paradigms agree within 10%, from 5.30 (LQR-BIC) to 5.82 (NGBoost). Under crisis the gap widens. Linear methods give 18.21–18.45 and the tree-based group 22.85–24.77. We group NGBoost with QRF and LGBM-Q throughout. Its estimation target is distributional, but its base learners are regression trees (Section 3.2), so it inherits the same extrapolation limits. The crisis-to-non-crisis CRPS ratio sits at 3.2–3.5 for the linear methods and 4.0–4.5 for the tree methods.

Table 5: Mean CRPS by regime. Non-crisis: Sep 2018, Dec 2019, Mar 2021, Mar 2024. Crisis: Sep 2021, Jun 2022.

Model	Non-Crisis	Crisis
Naive	9.05	36.05
QR	5.66	18.21
LQR-BIC	5.30	18.45
QRF	5.65	22.85
LGBM-Q	5.56	24.77
NGBoost	5.82	23.30

Diebold–Mariano tests confirm the gap is significant, but only at the upward-shift anchor. At September 2021 both linear methods beat every tree method, with all $p < 0.001$ and DM statistics from -8.7 to -10.4 . June 2022 does not follow the same pattern. Only one of the six linear–tree comparisons reaches significance there, and it favors the tree method (LQR-BIC against LGBM-Q, $p = 0.047$), so the linear group does not hold a consistent edge. The linear advantage under crisis therefore traces to the upward shift at September 2021 and not to crisis conditions across the board. This fits the mechanism described below. Tree methods pay a penalty only when the test window moves past the training range, which does not happen at June 2022 because the training window already covers the crisis.

Figure 2 shows the same split in coverage terms. September 2021 is the only anchor where the paradigms separate completely. Linear methods hold PICP-90 between 0.85 and 0.90, while every tree-based method falls to 0.46–0.58. Their intervals are correspondingly wider, with PINAW-90 at 0.34–0.36 against 0.29–0.31. This measures the extrapolation asymmetry directly. Linear quantiles widen through the fitted coefficients as the price level moves, whereas tree quantiles stay bounded by the training response range.

Table 6 reports the CQR-induced percentage change in CRPS per anchor and base model. The effect of CQR depends sharply on the paradigm and on whether $\mathcal{T}_{\text{cal}}$ and $\mathcal{T}_{\text{test}}$ fall in different regimes. September 2021, the canonical regime-shift anchor, shows the sharpest split. All three tree-based methods cluster at +40 to +68% degradation (QRF +54.7%, LGBM-Q +40.2%, NGBoost +67.7%), while *both* linear methods *benefit* (−3.5% for QR, −5.1% for LQR-BIC). March 2024 is a milder downward shift, as post-crisis prices normalized. There the tree penalty persists at reduced magnitude (15–33%), and the linear methods move from benefit to mild penalty (+5% to +13%).

Diebold–Mariano tests confirm these degradations. All three tree methods reject at $p < 0.001$ at both September 2021 and March 2024. The milder linear penalties at March 2024 are significant as well (QR $p < 0.001$, LQR-BIC $p = 0.007$). At June 2022 every method except NGBoost rejects, and unregularized QR is the most severely affected ($p < 0.001$). Of the two linear improvements at September 2021, only LQR-BIC’s is significant ($p = 0.031$), not QR’s ($p = 0.32$).

June 2022 reveals a different mechanism. Its training window already covers the crisis, so by the regime-shift logic CQR damage should be modest, and for LQR-BIC and the tree-based group it is (+1.6% to +15.4%). For unregularized QR, however, CQR degrades CRPS by 52% and PICP-90 falls from 0.86 to 0.47. Here the damage comes from the other half of the conformalization step, the proper-train refit, not from the conformal correction, which is large and

⁶Christoffersen-90 p-values: median below machine precision, maximum 1.5×10^{-12} . Ljung–Box at lag 24: 66/66 pairs reject at $\alpha = 0.05$.

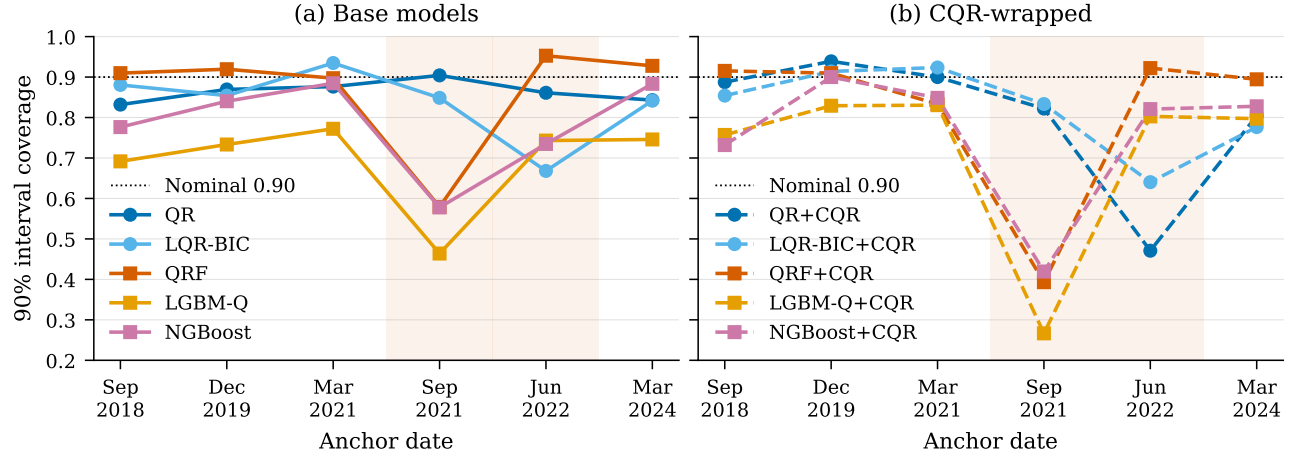


Figure 2: Empirical 90% coverage (PICP-90) across anchors for base models (left) and CQR-wrapped variants (right). The dotted line marks the nominal 0.90 target, and pink shading marks crisis anchors.

Table 6: CQR-induced percentage change in CRPS, shown per anchor and base model. Positive values indicate that CQR worsens CRPS. September 2021 exhibits the canonical paradigm split, with tree-based methods degrading by 40–68% and linear methods improving by 3.5–5%.

Anchor	QR	LQR-BIC	QRF	LGBM-Q	NGBost
Sep 2018	+3.2%	+7.5%	+11.0%	+19.5%	+7.3%
Dec 2019	−3.1%	−1.3%	+1.3%	−0.7%	−4.2%
Mar 2021	+1.0%	+0.3%	+9.4%	+14.2%	+9.0%
Sep 2021	−3.5%	−5.1%	+54.7%	+40.2%	+67.7%
Jun 2022	+52.0%	+15.4%	+5.5%	+11.2%	+1.6%
Mar 2024	+5.1%	+12.6%	+33.4%	+25.7%	+15.3%

positive (+42.5 EUR/MWh at the 90% level). Dropping the last 15% of a high-volatility training window shifts the refit QR’s quantile function about 25 EUR/MWh below the full-training fit over the test window and destabilizes its outer quantiles, raising the share of test hours needing monotone rearrangement from 31% to 68%. A symmetric widening cannot repair a location error, so the wider intervals still under-cover. LQR-BIC’s ℓ_1 regularization anchors the refit, whose median shift is a third of QR’s, limiting damage to +15%.

LQR-BIC achieves lower mean base CRPS than QR (9.68 versus 9.85), with per-anchor differences spanning ± 0.92 EUR/MWh and LQR-BIC favored at five of six anchors. Under conformalization it achieves 10.41 versus QR+CQR’s 11.98 (−13%), driven primarily by its robustness at June 2022. Regularization in the base model thus translates into more robust CQR behavior, particularly when the calibration set itself sits in a high-volatility regime.

The mechanism is clearest at September 2021. The training window rarely exceeded 100 EUR/MWh (5.1% of hours, concentrated in the calibration tail), while the test window spanned 60–248 EUR/MWh. The conformity-score distribution on $\mathcal{T}_{\text{cal}}$, the last 15% of training and so drawn from the calmer pre-shift regime, is unrepresentative of $\mathcal{T}_{\text{test}}$, which leaves the single conformal correction biased against the high-price regime. Tree methods cannot

extrapolate beyond the training response range, so they suffer most. NGBost degrades most of all, despite its distributional estimation target, and this points to the base learner’s extrapolation behavior as the binding factor rather than the form of the predictive distribution. Linear methods extrapolate through fitted coefficients and absorb the shift through the separately calibrated load and renewable forecasts. Figures 3 and 4 show the per-anchor CRPS surface and the ΔCRPS heatmap, whose “hot” rows are the visual signature of these mechanisms.

Tree methods amplify this because their residual scale responds strongly to the local distribution of training data. A linear method’s extrapolation error is bounded by coefficient mis-specification, whereas the tree-method error under a shift includes a *floor*, the maximum observed training price, that does not shrink with sample size, and the CQR correction estimated on the same residuals inherits this bias. Our finding is complementary to the non-exchangeable conformal results [2, 32, 33] reviewed in Section 2: those concern how the marginal coverage guarantee degrades under shift, whereas we identify a systematic *paradigm*-dependence of CRPS cost at a fixed level of exchangeability violation. An online adaptive scheme [33] combined with a base-paradigm-aware choice of base model and a regime-stratified correction is a natural direction. As a preliminary probe of the first ingredient, we ran a daily-batched adaptive conformal inference variant [32] on the same six-anchor protocol, updating the miscoverage level once per delivery day from the previous day’s realized coverage ($\gamma = 0.05$), with base model and calibration scores held fixed. At September 2021 it repairs only about 13% of the tree-method degradation (+54% under CQR versus +47% adaptive, averaged over the three tree methods), and the 90% correction saturates at the largest calibration score on 14 to 20 of the 30 test days, driving the effective level below zero without reaching nominal coverage. The paradigm split is unchanged: linear methods still improve slightly (−3.5 to −5.5%) while tree methods remain severely degraded. The binding constraint under regime shift is therefore the calibration *score set*, held in the pre-shift regime by the frozen protocol, not the calibration level that online adaptation corrects, which motivates methods

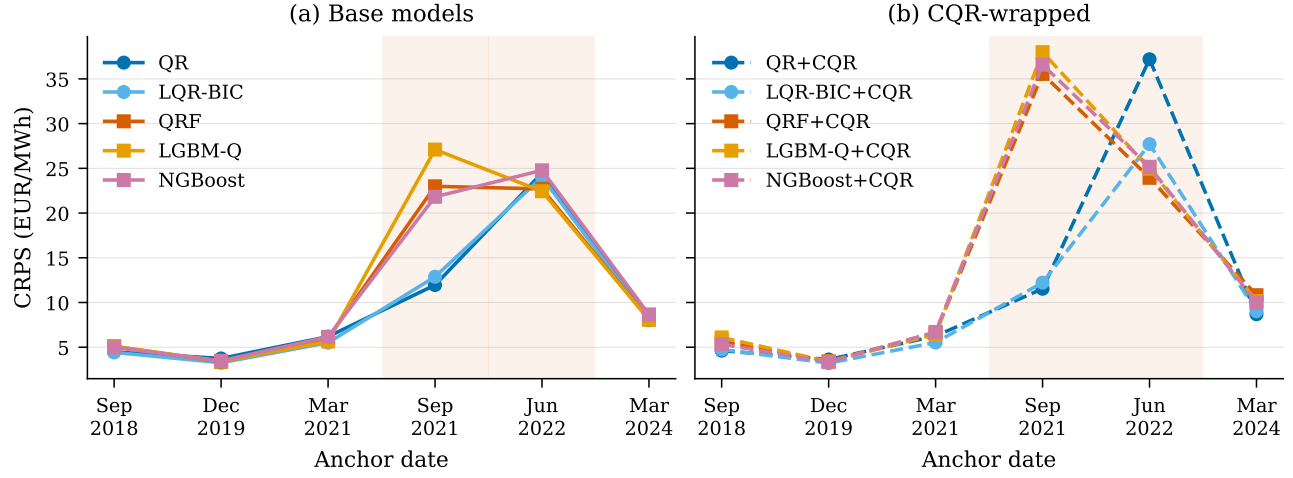


Figure 3: Per-anchor CRPS (EUR/MWh) for the five base models (left) and their CQR-wrapped counterparts (right). Pink shading marks crisis anchors. At Sep 2021 tree methods (red/orange) ramp up sharply under CQR while linear methods (blue) remain controlled or improve.

adapting the scores themselves. A full treatment, including step-size sensitivity and rolling recalibration, is deferred to future work.

5.3 Practical and Operational Implications

Three recommendations follow. First, paradigm choice matters little under stable conditions, so a linear method is a reasonable default on computational grounds. Linear methods fit and tune two to three orders of magnitude faster, and LQR-BIC is preferred where its small extra cost over QR is acceptable. Second, under regime-shift risk, linear quantile methods are markedly more robust than tree-based ones. Third, CQR should not be applied uniformly when the calibration set and test window straddle a regime boundary. In that case, either prefer a regularized linear base model or stratify the correction by regime.

These findings rest on a single market, the Dutch EPEX day-ahead auction, whose price formation, renewable penetration, and interconnection differ from other markets. The neighboring German market clears through the same mechanism and is directly interconnected, so we expect the qualitative findings to transfer. Whether they hold quantitatively where volatility and negative-price dynamics differ is an empirical question, and multi-market validation is a priority for future work. The regime-shift anchors are where probabilistic forecasts matter most downstream. Battery dispatch, demand-response triggering, and short-term hedging all rely on interval widths to size positions and manage collateral, so under-coverage translates into realized losses and over-coverage into foregone profit. The September 2021 degradation implies that a naive “wrap your model in CQR” default would yield forecasts with 1.4 to 1.7 times the base model’s CRPS exactly when reliable intervals matter most. For utilities, grid operators, and traders, a conformal wrapper is not a substitute for regime-aware methodology. Either the base model must be regime-robust, which favors the linear paradigm under structural-break risk, or the conformalization step must be stratified. Two further recommendations follow. A regularized linear base model gives better CQR robustness than its unregularized counterpart. And CQR damage is not confined

to calibration-to-test regime mismatch, since the proper-train refit inside the CQR construction can itself destabilize an unregularized linear base model when the training window is volatile.

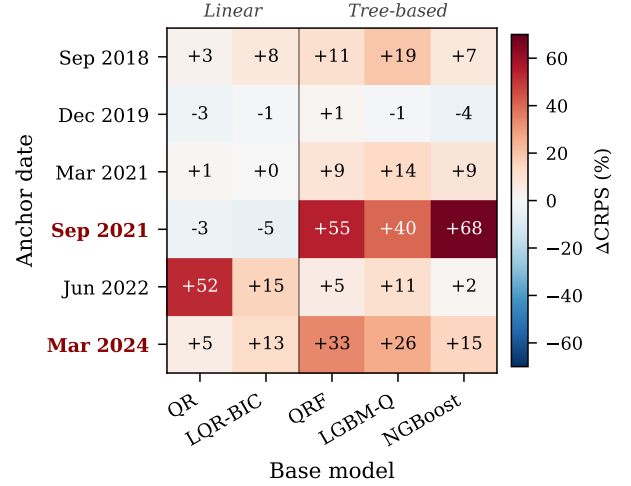


Figure 4: CQR-induced percentage change in CRPS for each anchor and base model. Positive values (red) indicate that CQR worsens CRPS. The “hot” September 2021 row for QRF, LGBM-Q, and NGBBoost marks the canonical regime-shift anchor, whereas the QR cell for June 2022 illustrates the complementary refit-sensitivity mechanism specific to unregularized linear methods.

5.4 Limitations

Three limitations remain. First, the fit-once-per-anchor protocol departs from the daily rolling re-estimation standard in operational EPF [15, 28, 35]. That departure is deliberate (Section 4.2), but the effect magnitudes reported here are specific to the frozen protocol. Quantifying how much of the paradigm-conditional effect survives once the calibration set tracks the regime is the central robustness question for future work. Our adaptive-conformal probe already indicates that adapting the calibration *level* alone, without refreshing

the scores, is insufficient. Second, the regime classification of the September 2021 and March 2024 anchors is ex-post. Pre-hoc detection (change-point analysis on rolling volatility, or break-detection from the structural-change econometrics literature) would be required for the operational regime-stratified scheme our findings indicate, and is left to follow-on work. Third, only the Normal parameterization of NGBoost is considered. Day-ahead prices are heavy-tailed and spike during regime shifts, exactly where a Normal underestimates tail mass, so a heavier-tailed family such as Student- t could change the result. A systematic comparison of parametric families, including distributional neural networks [18], is left to future work.

6 Conclusions and Future Work

Across six anchor dates on the Dutch EPEX day-ahead market, CQR's effect on CRPS proved paradigm- and regime-conditional: the five methods agree in calm regimes, linear methods lead in crisis, and at the September 2021 regime shift tree-based models degrade by 40–68% while linear ones improve. Regularization also matters under conformalization, LQR-BIC+CQR achieving 13% lower mean CRPS than QR+CQR. We traced these to the differing residual-scale behavior of tree versus linear base models and to the calibration set's location relative to the test regime. Future work has three threads. First, a pre-deployment diagnostic for when CQR is contraindicated under regime shift, together with a per-regime coverage guarantee via Mondrian conformal prediction [29]. Second, a regime-stratified CQR scheme (RS-CQR) that computes per-regime corrections from within-regime calibration scores, validated on the same six-anchor evaluation against plain CQR on CRPS and coverage. Third, an ensemble of paradigms with regime-conditional weights, combining linear robustness with tree-based flexibility, for which a stacking meta-learner is a natural candidate.

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